# ArmBench in the robot-policy adaptation landscape

Runtime reliability and statistical evidence are distinct contributions from supervised fine-tuning, RL adaptation, and human-in-the-loop learning.

| METHOD / STATUS | ADAPTATION LOCUS | LEARNING SIGNAL | LATENCY QUESTION | EVIDENCE DOMAIN |
|---|---|---|---|---|
| **ArmBench**<br>PROJECT EVIDENCE<br>NOT PEER REVIEWED | Runtime dispatcher<br>Frozen official pi0.5 | No training | Measure response age;<br>select suffix or fail closed | LIBERO simulation<br>4 suites / 40 tasks<br>paired + manifest-bound |
| **RTC**<br>NEURIPS 2025<br>FORMAL PAPER | Inference-time<br>action-chunk inpainting | No retraining | Freeze committed actions;<br>inpaint the continuation | Dynamic simulation +<br>real bimanual tasks |
| **OpenVLA-OFT**<br>RSS 2025<br>FORMAL PAPER | VLA weight<br>fine-tuning | Offline imitation /<br>L1 regression | Faster parallel decoding;<br>async staleness not central | LIBERO simulation +<br>real bimanual ALOHA |
| **DPPO**<br>ICLR 2025<br>FORMAL PAPER | Diffusion-policy<br>RL fine-tuning | Policy gradients /<br>reward | Not an asynchronous<br>runtime method | Simulation + zero-shot<br>hardware deployment |
| **HIL-SERL**<br>SCIENCE ROBOTICS 2025<br>FORMAL PAPER | Real-world policy<br>learning | Demos + human<br>corrections + RL | Not an action-chunk<br>latency method | Real robot +<br>dexterous tasks |

Solid border: formal venue/journal paper    Dashed border: project evidence, not a publication

### ARMBENCH TODAY

**Strong systems evidence, limited policy-learning claim**

- Frozen official pi0.5 checkpoint
- No fine-tuning or hidden task tuning
- Frozen protocols + matched conditions
- Exact tests, multiplicity, bootstrap CI
- Source/checkpoint/video attestation

### TOP-CONFERENCE GAP

- Independent inference / control clocks
- Same-checkpoint head-to-head with RTC
- More than one VLA checkpoint
- Real-robot closed-loop validation
- Mechanism beyond fixed suffix skipping

These gaps test generality and deployment realism; they do not imply that RL is mandatory.

### NEAREST SCIENTIFIC COMPARATOR

RTC: training-free, asynchronous action-chunk execution with simulation and real-robot evidence.

Status checked 2026-08-05. [1] OpenVLA-OFT: RSS 2025, DOI 10.15607/RSS.2025.XXI.017. [2] DPPO: ICLR 2025, arXiv:2409.00588. [3] HIL-SERL: Science Robotics 2025, DOI 10.1126/scirobotics.ads5033. [4] RTC: NeurIPS 2025, arXiv:2506.07339. ArmBench scope follows the repository's frozen evidence; see the adjacent source note.